

Task 5 – Librem5 Battery Optimisation

1. Calculate the time it is taking to drain from 100 to 0:

This report is written on the basis of observations done physically on the device:

The device was at 100% at 12:04 AM 27-03-25

Then the device was left idle completely until full drain. The rest observations:

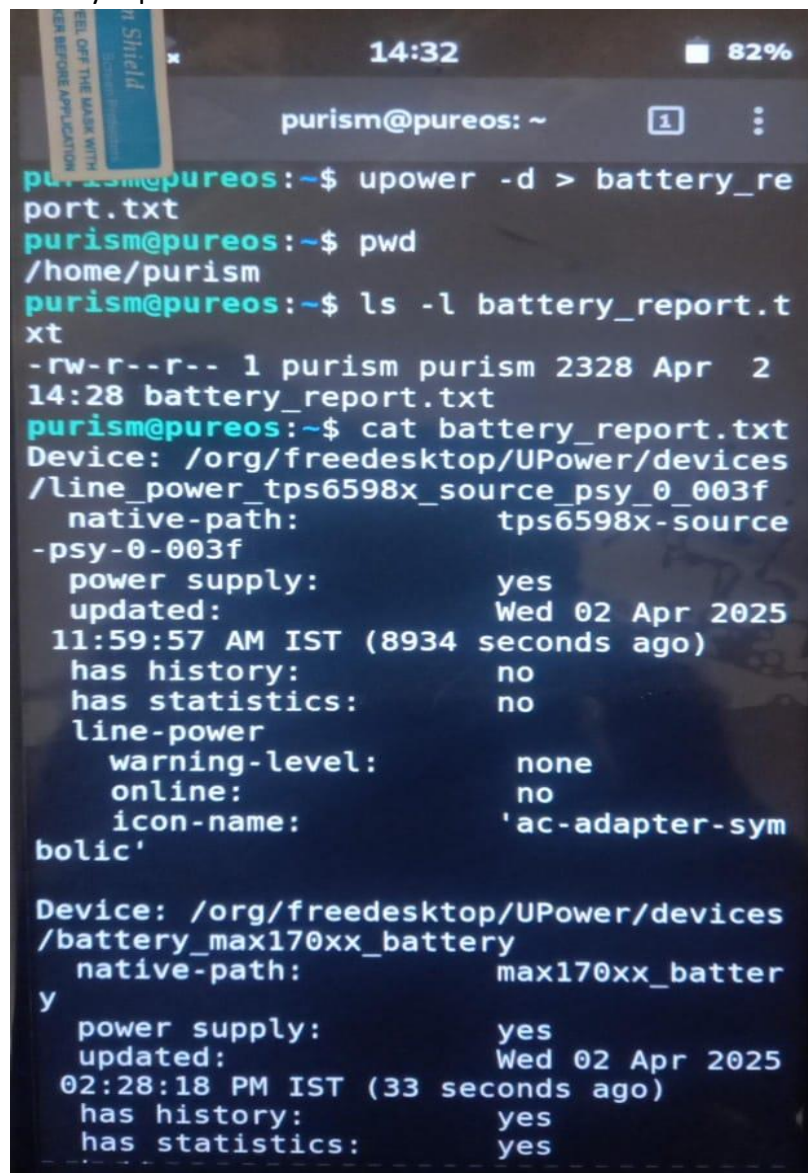
72% at 6:31 AM.

65% at 8:02 AM.

26% at 4:30 PM

And completely dead at around 6:30 PM.

Battery report:



The screenshot shows a terminal window on a Librem5 device. The status bar at the top indicates the time is 14:32 and the battery level is 82%. The terminal prompt is `purism@pureos: ~`. The user has executed the following commands and received the following output:

```
purism@pureos:~$ upower -d > battery_report.txt
purism@pureos:~$ pwd
/home/purism
purism@pureos:~$ ls -l battery_report.txt
-rw-r--r-- 1 purism purism 2328 Apr  2 14:28 battery_report.txt
purism@pureos:~$ cat battery_report.txt
Device: /org/freedesktop/UPower/devices/line_power_tps6598x_source_psy_0_003f
  native-path: tps6598x-source-psy-0-003f
  power supply: yes
  updated: Wed 02 Apr 2025 11:59:57 AM IST (8934 seconds ago)
  has history: no
  has statistics: no
  line-power
    warning-level: none
    online: no
    icon-name: 'ac-adapter-symbolic'

Device: /org/freedesktop/UPower/devices/battery_max170xx_battery
  native-path: max170xx_battery
  power supply: yes
  updated: Wed 02 Apr 2025 02:28:18 PM IST (33 seconds ago)
  has history: yes
  has statistics: yes
```

```

14:32 82%
purism@pureos: ~
02:28:18 PM IST (33 seconds ago)
has history: yes
has statistics: yes
battery
  present: yes
  rechargeable: yes
  state: discharging
  warning-level: none
  energy: 13.7187 Wh
  energy-empty: 0 Wh
  energy-full: 16.4604 Wh
  energy-full-design: 13.8572 Wh
  energy-rate: 1.77236 W
  voltage: 3.89156 V
  time to empty: 7.7 hours
  percentage: 83%
  temperature: 32.3 degrees C
  capacity: 100%
  technology: lithium-ion
  icon-name: 'battery-full-s
symbolic'
History (rate):
  1743584298 1.772 discharging
Device: /org/freedesktop/UPower/devices
/line_power_bq25890_charger
  native-path: bq25890-charger
  power supply: yes
  updated: Wed 02 Apr 2025
11:59:58 AM IST (8933 seconds ago)
has history: no
has statistics: no
line-power

```

```

14:32 82%
purism@pureos: ~
has statistics: no
line-power
  warning-level: none
  online: no
  icon-name: 'ac-adapter-sym
symbolic'
Device: /org/freedesktop/UPower/devices
/DisplayDevice
  power supply: yes
  updated: Wed 02 Apr 2025
02:28:18 PM IST (33 seconds ago)
has history: no
has statistics: no
battery
  present: yes
  state: discharging
  warning-level: none
  energy: 13.7187 Wh
  energy-full: 16.4604 Wh
  energy-rate: 1.77236 W
  time to empty: 7.7 hours
  percentage: 83%
  icon-name: 'battery-full-s
symbolic'
Daemon:
  daemon-version: 0.99.11
  on-battery: yes
  lid-is-closed: no
  lid-is-present: no
  critical-action: PowerOff
purism@pureos:~$

```

2. Find the processes which are running consistently:

2.1 List all running processes sorted by CPU usage:

```
14:34 82%
purism@pureos: ~
critical-action: PowerOff
purism@pureos:~$ ps aux --sort=-%cpu |
head -20
USER          PID %CPU %MEM    VSZ   RSS
TTY          STAT START   TIME COMMAND
purism        4580  4.4   1.7 557888 53576
?            Rl    13:18   3:23 /usr/bin/kg
x --gaplication-service
purism        4178  1.6  12.0 2879580 3658
76 tty7      Sl+   13:11   1:22 /usr/lib/fi
refox-esr/firefox-esr
purism        4368  0.3   5.0 2470196 1533
64 tty7      Sl+   13:12   0:17 /usr/lib/fi
refox-esr/firefox-esr -contentproc -chi
ldID 3 -isForBrowser -prefsLen 32781 -p
refMapSize 217952 -jsInitLen 277276 -pa
rentBuildID 20230529170948 -appDir /usr
/lib/firefox-esr/browser 4178 true tab
root         527  0.2   0.0   2152   1152
?           Ss    Apr01   4:27 /usr/sbin/j
itterentropy-rngd
message+     564  0.1   0.1  10452   5120
?           Ss    Apr01   1:47 /usr/bin/db
us-daemon --system --address=systemd: -
-nofork --nopidfile --systemd-activatio
n --syslog-only
root         565  0.1   0.5 262368 17924
?           Ssl   Apr01   2:02 /usr/sbin/N
etworkManager --no-daemon
purism        723  0.1   3.2 874008 99564
tty7        Ssl+  Apr01   2:28 /usr/bin/ph
oc -S -C /usr/share/phosh/phoc.ini -E b
ash -lc 'gnome-session --disable-accele
ration-check --session=phosh --builtin'
```

```
14:35 82%
purism@pureos: ~
etwanager --no-daemon
purism        723  0.1   3.2 874008 99564
tty7        Ssl+  Apr01   2:28 /usr/bin/ph
oc -S -C /usr/share/phosh/phoc.ini -E b
ash -lc 'gnome-session --disable-accele
ration-check --session=phosh --builtin'
purism        760  0.1   0.1 230176  4864
?           Ssl   Apr01   1:59 /usr/bin/gt
hd
purism        949  0.1   5.8 1102232 1785
08 tty7      Sl+   Apr01   3:23 /usr/libexe
c/phosh
root         2638  0.1   0.3  10520   9264
?           S<Ls 00:00   1:02 /usr/bin/at
op -R -w /var/log/atop/atop_20250402 60
0
root         1  0.0   0.3 165120 11576
?           Ss    Apr01   0:13 /sbin/init
splash
root         2  0.0   0.0     0     0
?           S    Apr01   0:01 [kthreadd]
root         3  0.0   0.0     0     0
?           I<   Apr01   0:00 [rcu_gp]
root         4  0.0   0.0     0     0
?           I<   Apr01   0:00 [rcu_par_gp]
]
root         5  0.0   0.0     0     0
?           I<   Apr01   0:00 [slub_flush
wq]
root         6  0.0   0.0     0     0
?           I<   Apr01   0:00 [netns]
root        10  0.0   0.0     0     0
?           I<   Apr01   0:00 [mm_percpu_
wal
```

```

14:35 82%
purism@pureos: ~
nc
purism 949 0.1 5.8 1102232 1785
08 tty7 Sl+ Apr01 3:23 /usr/libexe
c/phosh
root 2638 0.1 0.3 10520 9264
? S<Ls 00:00 1:02 /usr/bin/at
op -R -w /var/log/atop/atop_20250402 60
0
root 1 0.0 0.3 165120 11576
? Ss Apr01 0:13 /sbin/init
splash
root 2 0.0 0.0 0 0
? S Apr01 0:01 [kthreadd]
root 3 0.0 0.0 0 0
? I< Apr01 0:00 [rcu_gp]
root 4 0.0 0.0 0 0
? I< Apr01 0:00 [rcu_par_gp]
]
root 5 0.0 0.0 0 0
? I< Apr01 0:00 [slub_flush
wq]
root 6 0.0 0.0 0 0
? I< Apr01 0:00 [netns]
root 10 0.0 0.0 0 0
? I< Apr01 0:00 [mm_percpu_
wq]
root 11 0.0 0.0 0 0
? I Apr01 0:00 [rcu_tasks_
kthreadd]
root 12 0.0 0.0 0 0
? I Apr01 0:00 [rcu_tasks_
rude_kthreadd]
purism@pureos:~$

```

2.2 Check which processes have been running the longest:

```

14:37 81%
purism@pureos: ~
I Apr01 0:00 [rcu_tasks_
rude_kthreadd]
purism@pureos:~$ ps -eo pid,etimes,cmd
--sort=etimes | head -20
PID ELAPSED CMD
5217 0 ps -eo pid,etimes,cmd -
-sort=etimes
5218 0 head -20
5213 83 [kworker/2:2-events]
5212 95 [kworker/0:0H-mmc_compl
ete]
5211 126 [kworker/0:0-events]
5206 247 [kworker/3:2-kdmflush/2
54:0]
5186 467 [kworker/2:0-events]
5182 474 [kworker/0:2H-mmc_compl
ete]
5178 478 /usr/libexec/gvfsd-meta
data
5149 497 [kworker/3:2H]
5138 536 [kworker/u8:0-events_un
bound]
5136 541 [kworker/0:2-rcu_gp]
5137 541 [kworker/1:0-events]
5126 956 [kworker/0:1-pm]
5101 1461 [kworker/u8:3-kcryptd/2
54:0]
5072 1604 [kworker/u8:2+events_un
bound]
5073 1604 [kworker/2:0H]
5069 1631 [kworker/3:0-events]
5030 3091 [kworker/1:0H]
purism@pureos:~$

```

Htop:

```
14:38 81%
purism@pureos: ~

0[| 5.8%] Tasks: 110, 39
1[| 0.6%] Load average:
2[| 0.0%] Uptime: 1 day,
3[| 3.2%]
Mem[1.28G/2.9]
Swp[ 0K/1.45G]

  PID USER      PRI  NI  VIRT   RES   SHR S
 4580 purism    20   0  545M  54668  3840 R
 5223 purism    20   0  15148  3840  3840 R
    1 root      20   0   161M  11576  11576 R
  434 root      20   0  94940  54016  54016 R
  447 root      20   0  22092  6528  6528 R
  453 systemd-n 20   0  16752  6400  6400 R
  527 root      20   0   2152  1152  1152 R
  528 systemd-r 20   0  24068  11716  11716 R
  529 systemd-t 20   0  88240  6016  6016 R
  547 root      20   0   238M  8456  8456 R
  549 avahi    20   0   8048  3072  3072 R
  551 root      0  -20   1944  1152  1152 R
  553 root      20   0   238M  8456  8456 R
  557 root      20   0  13692  2304  2304 R
  564 messagebu 20   0  10452  5120  5120 R
  565 root      20   0   256M  17924  17924 R
  569 root      20   0  1123M  14064  14064 R
  572 root      20   0   231M  11092  11092 R
  575 root      20   0  31052  6144  6144 R
  576 root      20   0   231M  11092  11092 R
  580 root      20   0  1123M  14064  14064 R
  581 root      20   0   383M  16668  16668 R
  583 root      20   0  1123M  14064  14064 R

F1Help F2Setup F3Search F4Filter F5Tree
```

3. Identify what each process is doing:

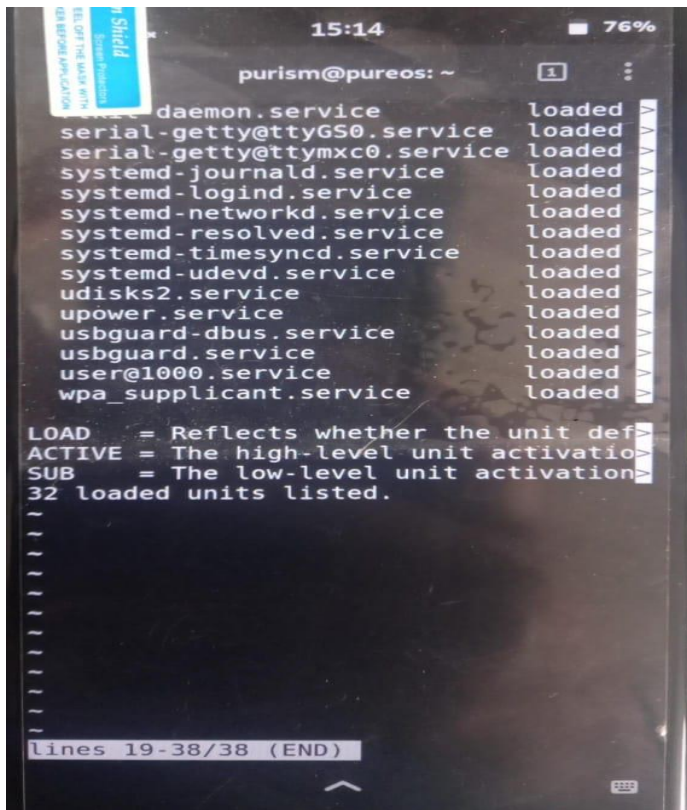
```
14:50 79%
purism@pureos: ~

478 /usr/libexec/gvfsd-metadata
data
5149 497 [kworker/3:2H]
5138 536 [kworker/u8:0-events_un
bound]
5136 541 [kworker/0:2-rcu_gp]
5137 541 [kworker/1:0-events]
5126 956 [kworker/0:1-pm]
5101 1461 [kworker/u8:3-kcryptd/2
54:0]
5072 1604 [kworker/u8:2+events_un
bound]
5073 1604 [kworker/2:0H]
5069 1631 [kworker/3:0-events]
5030 3091 [kworker/1:0H]
purism@pureos:~$ htop
purism@pureos:~$ ps -p 5217 -o pid,user
,%cpu,%mem,etime,cmd
  PID USER      %CPU %MEM    ELAPSED
  CMD
purism@pureos:~$ ps -p 5178 -o pid,user
,%cpu,%mem,etime,cmd
  PID USER      %CPU %MEM    ELAPSED
  CMD
  5178 purism    0.0  0.2      19:14
  /usr/libexec/gvfsd-metadata
purism@pureos:~$ ps -p 5030 -o pid,user
,%cpu,%mem,etime,cmd
  PID USER      %CPU %MEM    ELAPSED
  CMD
  5030 root      0.0  0.0      01:04:43
  [kworker/1:0H]
purism@pureos:~$
```

4. List active services and processes:

```
15:12 76%
purism@pureos: ~
purism@pureos:~$ systemctl list-units -
-t=service --state=running
UNIT                                LOAD
accounts-daemon.service            loaded
atop.service                        loaded
atopacct.service                   loaded
avahi-daemon.service               loaded
bluetooth-brcmfmac.service         loaded
bluetooth.service                  loaded
colord.service                      loaded
cron.service                       loaded
dbus.service                       loaded
gnss-share.service                 loaded
iio-sensor-proxy.service           loaded
jitterentropy.service              loaded
ModemManager.service              loaded
NetworkManager.service            loaded
packagekit.service                loaded
phosh.service                      loaded
polkit.service                     loaded
rtkit-daemon.service               loaded
serial-getty@ttyGS0.service        loaded
lines 1-20...skipping...
UNIT                                LOAD
accounts-daemon.service            loaded
atop.service                        loaded
atopacct.service                   loaded
avahi-daemon.service               loaded
bluetooth-brcmfmac.service         loaded
bluetooth.service                  loaded
colord.service                      loaded
cron.service                       loaded
dbus.service                       loaded
```

```
15:13 76%
purism@pureos: ~
accounts-daemon.service            loaded
atop.service                        loaded
atopacct.service                   loaded
avahi-daemon.service               loaded
bluetooth-brcmfmac.service         loaded
bluetooth.service                  loaded
colord.service                      loaded
cron.service                       loaded
dbus.service                       loaded
gnss-share.service                 loaded
iio-sensor-proxy.service           loaded
jitterentropy.service              loaded
ModemManager.service              loaded
NetworkManager.service            loaded
packagekit.service                loaded
phosh.service                      loaded
polkit.service                     loaded
rtkit-daemon.service               loaded
serial-getty@ttyGS0.service        loaded
serial-getty@ttymxc0.service       loaded
systemd-journald.service           loaded
systemd-logind.service             loaded
systemd-networkd.service           loaded
systemd-resolved.service           loaded
systemd-timesyncd.service          loaded
systemd-udevd.service              loaded
udisks2.service                   loaded
upower.service                     loaded
usbguard-dbus.service              loaded
usbguard.service                  loaded
user@1000.service                  loaded
lines 1-32
```

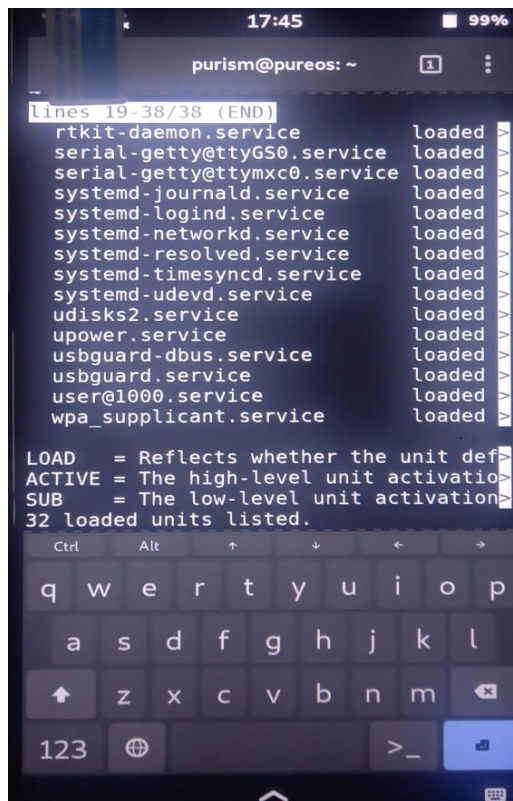


5. Optimize the battery life:

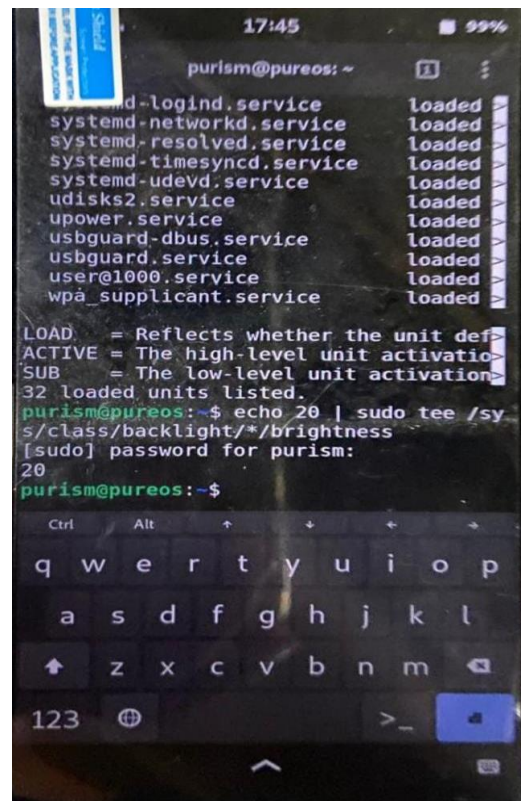
5.1 Reduce Display Power Usage:

- Lower brightness manually:

Before:



After:



- Reduce screen timeout:

A terminal window on a device with a 'Shield' sticker. The terminal shows a list of loaded system services, followed by explanatory text for LOAD, ACTIVE, and SUB. Then, the user runs a command to echo '20' to a file, followed by a password prompt and the command to set the idle delay to 30 seconds using gsettings.

```

17:48 98%
purism@pureos: ~
systemd-resolved.service loaded
systemd-timesyncd.service loaded
systemd-udevd.service loaded
udisks2.service loaded
upower.service loaded
usbguard-dbus.service loaded
usbguard.service loaded
user@1000.service loaded
wpa_supplicant.service loaded

LOAD = Reflects whether the unit def
ACTIVE = The high-level unit activatio
SUB = The low-level unit activation
32 loaded units listed.
purism@pureos:~$ echo 20 | sudo tee /sys/class/backlight/*/brightness
[sudo] password for purism:
20
purism@pureos:~$ gsettings set org.gnome.desktop.session idle-delay 30
purism@pureos:~$

```

5.2 Disable Unused Hardware

- Turn off Wi-Fi when not needed:

A terminal window on a device with a 'Shield' sticker. The terminal shows the same list of loaded system services and explanatory text as the previous screenshot. The user then runs the command to turn off Wi-Fi using nmcli.

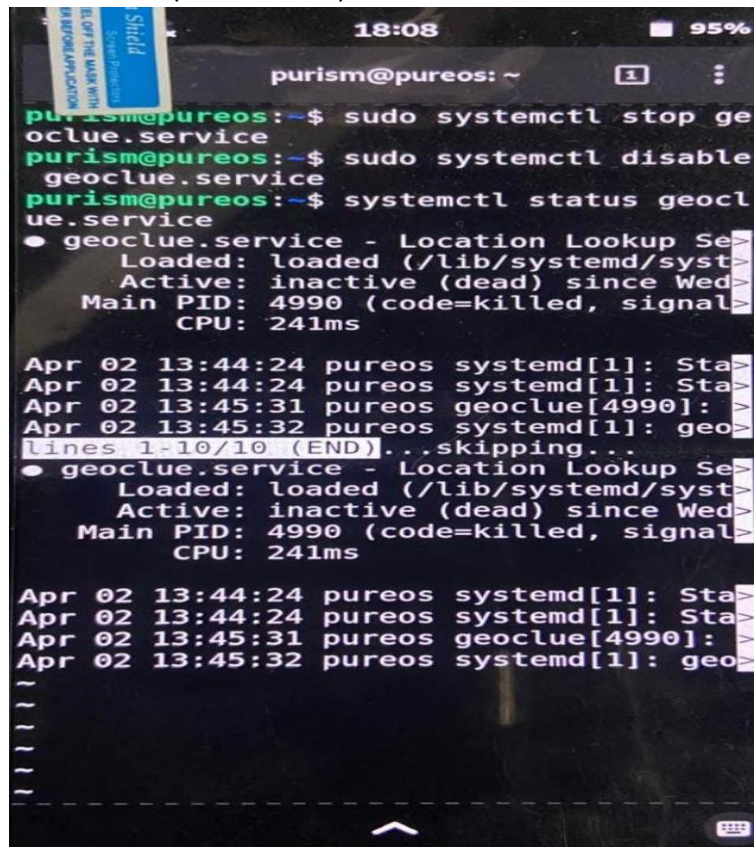
```

17:55 97%
purism@pureos: ~
systemd-timesyncd.service loaded
systemd-udevd.service loaded
udisks2.service loaded
upower.service loaded
usbguard-dbus.service loaded
usbguard.service loaded
user@1000.service loaded
wpa_supplicant.service loaded

LOAD = Reflects whether the unit def
ACTIVE = The high-level unit activatio
SUB = The low-level unit activation
32 loaded units listed.
purism@pureos:~$ echo 20 | sudo tee /sys/class/backlight/*/brightness
[sudo] password for purism:
20
purism@pureos:~$ gsettings set org.gnome.desktop.session idle-delay 30
purism@pureos:~$ nmcli radio wifi off
purism@pureos:~$

```

- Disable GPS (via software):

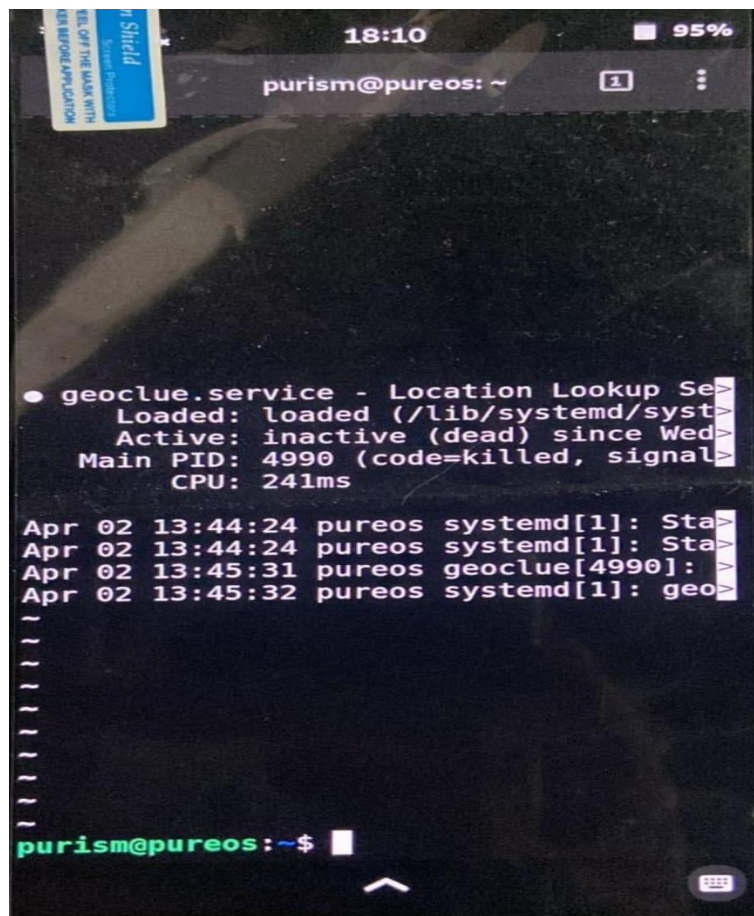


A screenshot of a terminal window on a PureOS device. The terminal shows the user 'purism@pureos' executing a series of commands to stop and disable the geoclue.service. The status of the service is shown as 'inactive (dead)'. The terminal also displays system logs for the systemd and geoclue processes.

```
purism@pureos:~$ sudo systemctl stop geoclue.service
purism@pureos:~$ sudo systemctl disable geoclue.service
purism@pureos:~$ systemctl status geoclue.service
● geoclue.service - Location Lookup Service
   Loaded: loaded (/lib/systemd/system-geoclue.service; vendor preset: enabled)
   Active: inactive (dead) since Wednesday, April 3, 2024 13:45:32 UTC
   Main PID: 4990 (code=killed, signal=SIGKILL)
   CPU: 241ms

Apr 02 13:44:24 pureos systemd[1]: Starting geoclue.service: Location Lookup Service...
Apr 02 13:44:24 pureos systemd[1]: Started geoclue.service: Location Lookup Service.
Apr 02 13:45:31 pureos geoclue[4990]: Shutting down...
Apr 02 13:45:32 pureos systemd[1]: geoclue.service: Main process exited, code=killed, signal=SIGKILL
lines 1-10/10 (END) ...skipping...
● geoclue.service - Location Lookup Service
   Loaded: loaded (/lib/systemd/system-geoclue.service; vendor preset: enabled)
   Active: inactive (dead) since Wednesday, April 3, 2024 13:45:32 UTC
   Main PID: 4990 (code=killed, signal=SIGKILL)
   CPU: 241ms

Apr 02 13:44:24 pureos systemd[1]: Starting geoclue.service: Location Lookup Service...
Apr 02 13:44:24 pureos systemd[1]: Started geoclue.service: Location Lookup Service.
Apr 02 13:45:31 pureos geoclue[4990]: Shutting down...
Apr 02 13:45:32 pureos systemd[1]: geoclue.service: Main process exited, code=killed, signal=SIGKILL
```



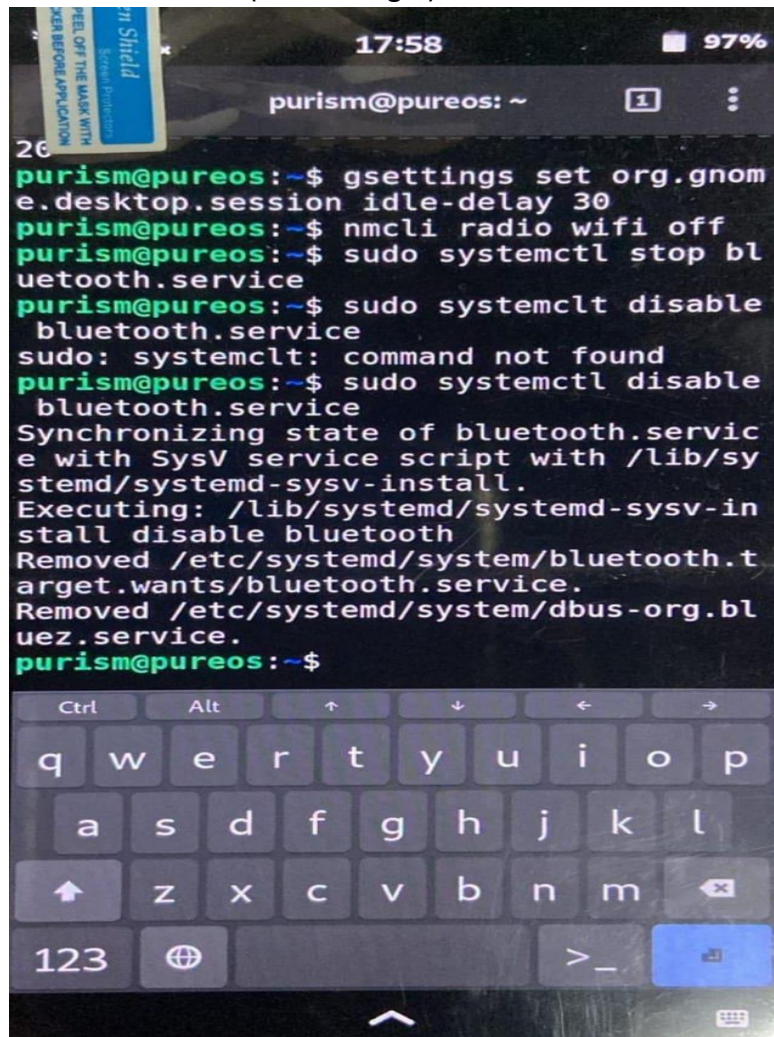
A second screenshot of the terminal window, showing the status of the geoclue.service after the previous commands. The service is confirmed to be 'inactive (dead)'. The terminal also displays system logs for the systemd and geoclue processes.

```
● geoclue.service - Location Lookup Service
   Loaded: loaded (/lib/systemd/system-geoclue.service; vendor preset: enabled)
   Active: inactive (dead) since Wednesday, April 3, 2024 13:45:32 UTC
   Main PID: 4990 (code=killed, signal=SIGKILL)
   CPU: 241ms

Apr 02 13:44:24 pureos systemd[1]: Starting geoclue.service: Location Lookup Service...
Apr 02 13:44:24 pureos systemd[1]: Started geoclue.service: Location Lookup Service.
Apr 02 13:45:31 pureos geoclue[4990]: Shutting down...
Apr 02 13:45:32 pureos systemd[1]: geoclue.service: Main process exited, code=killed, signal=SIGKILL

purism@pureos:~$
```

- Disable Bluetooth (if not using it):



```
purism@pureos: ~  
26  
purism@pureos:~$ gsettings set org.gnome.desktop.session idle-delay 30  
purism@pureos:~$ nmcli radio wifi off  
purism@pureos:~$ sudo systemctl stop bluetooth.service  
purism@pureos:~$ sudo systemctl disable bluetooth.service  
sudo: systemctl: command not found  
purism@pureos:~$ sudo systemctl disable bluetooth.service  
Synchronizing state of bluetooth.service with SysV service script with /lib/systemd/systemd-sysv-install.  
Executing: /lib/systemd/systemd-sysv-install disable bluetooth  
Removed /etc/systemd/system/bluetooth.target.wants/bluetooth.service.  
Removed /etc/systemd/system/dbus-org.bluez.service.  
purism@pureos:~$
```